DataFlock Project Documentation (Review)

1. Topic Selection

Project Title: Design and Development of ESG Data Collection and Management Platform (DataFlock).

2. Literature Review

ESG reports contain large volumes of unstructured information. Platforms like DataFlock help extract, validate, and store ESG data in structured databases using modern web technologies.

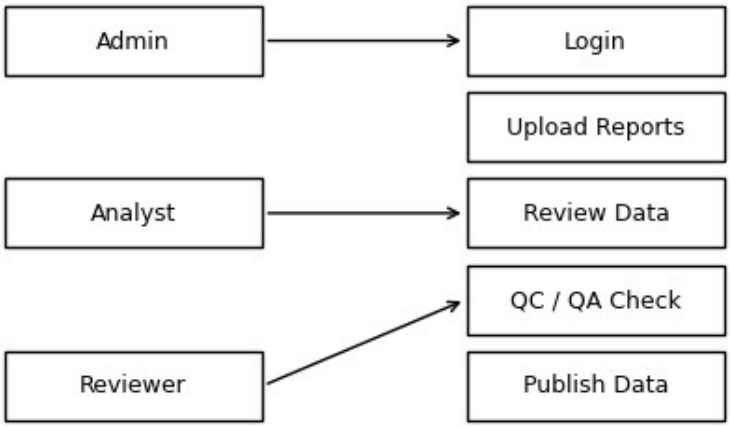
3. Environment Setup

Frontend: Angular
Backend: Node.js + Express.js
Database: MySQL
Tools: VS Code, Git

4. Initial Development

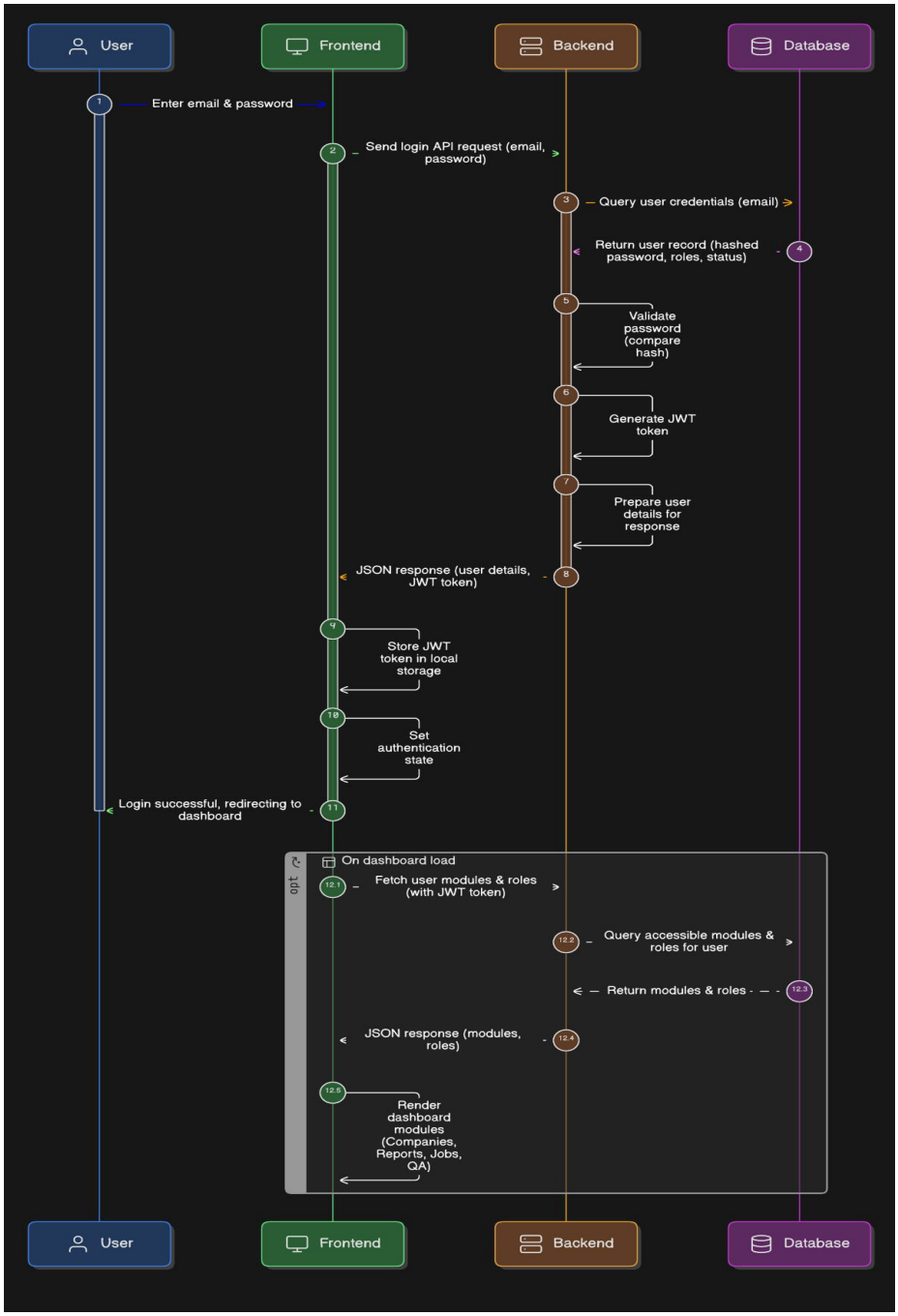
Modules implemented include authentication, company management, report upload, and ESG data storage tables.

5. Use Case Diagram

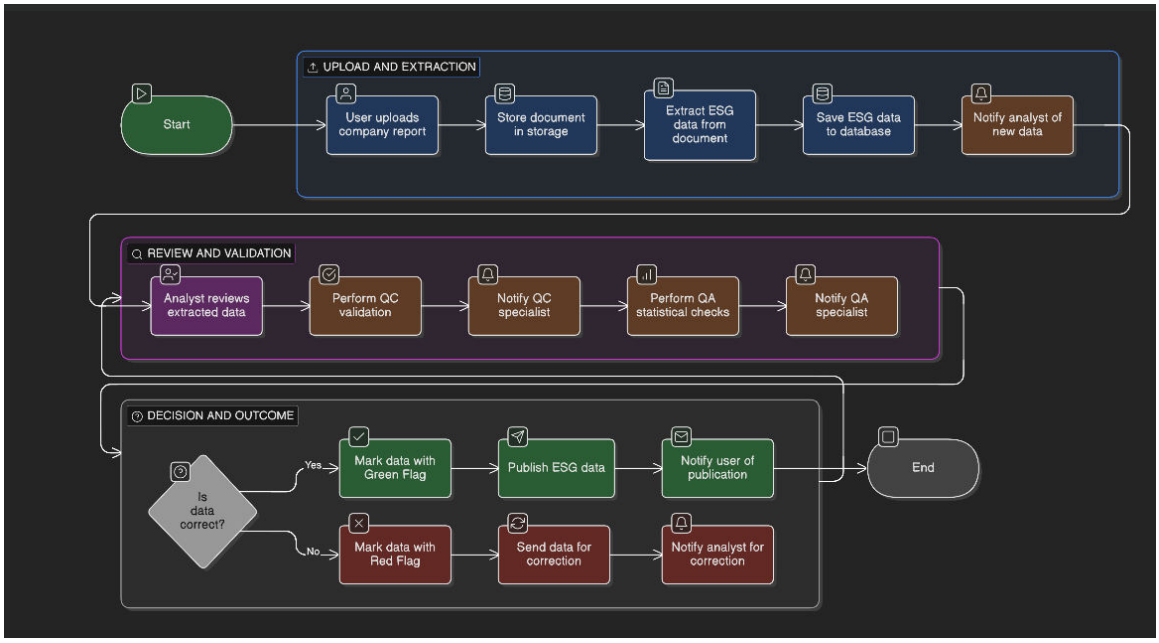


6. Class Diagram

7. Sequence Diagram



8. Activity Diagram



9. Table Design

Main tables: users, companies, company_reports, theme_state and ESG metric tables such as carbon_emission and energy_consumption.

10. Website Map

Login → Dashboard → Users → Companies → Jobs → QA → Data View